

Why do EUCAST have no systemic breakpoints for *Enterobacterales* with oral cephalosporins?

Version 2

There have been multiple questions from clinicians, particularly those working in orthopaedics, who have “successfully” used oral cephalosporins for prophylaxis and to treat *Enterobacterales* infections for many years”. They ask what has changed and why these agents are now considered inappropriate.

In EUCAST rationale documents it is stated that *Enterobacterales* are inappropriate targets in sites other than uncomplicated urinary tract infection, but there is no further explanation. EUCAST discussions were originally considers oral cephalosporins inappropriate for treatment of infections in other sites than the urinary tract infection for several reasons:

1. Comparison of free drug pharmacokinetics with MICs alone indicates that inadequate exposures are achieved for most agents or are borderline at best (see table).
2. The relevant pharmacodynamic relationship indicative of activity of cephalosporins is $fT > MIC$ and the target $\%fT > MIC$ is 40-50%. Approximate calculations based on common dosages indicate that activity is inadequate for all agents (see table). It should be emphasized that the figures in the table are based on pharmacokinetic parameter values for the mean of the population. Monte Carlo simulations would show that the $\%fT > MIC$ values are even less than those in the table for half the population treated.
3. Evidence of successful clinical use is anecdotal and may be unrelated to specific *Enterobacterales* isolates, which are rare in orthopaedic infection and often in mixtures of organisms both from bone and other sites.
4. Oral cephalosporins have mostly been used as “follow-up” therapy after successful parenteral treatment in hospital.

No studies have directly addressed the efficacy of oral cephalosporins in infections outside the lower urinary tract caused by *Enterobacterales*. The pharmacokinetics and efficacy of higher dosages leading to increased exposure, while theoretically possible for extra-urinary gram-negative infections for some of these agents, are unknown.

With regards to prophylaxis, there are no pharmacodynamic correlates but it is conventional to use agents that would be active against the relevant target pathogens in treatment.

PK and PD data for oral cephalosporins

Agent [Ref]	Common Dosage (g)	C _{max} (mg/L)	Protein binding (%)	t _{1/2} (h)	ECOFF <i>E. coli</i> (mg/L)	Approx. % fT>ECOFF
Cefaclor [1]	0.5 x 3	15.9 ± 5.7	50	0.69 ± .17	4	15
Cefadroxil [2]	0.5 x 2	17.9 ± 1.0	18-20	1.0-1.9	16	0
Cefalexin [3]	0.5 x 2	18.1 ± 4.3	10-20	1.5 ± 0.6	16	0
Cefixime [1,4]	0.4 x 1	2.5-6.1	60-70	2.7-4.6	2	10
Cefpodoxime [5]	0.2 x 2	1.8-2.6	22-33	3.3-4.2	2	0
Ceftibuten [1,6]	0.4 x 1	15.0 ± 3.3	63	2.5 ± 0.2	1	37
Cefuroxime-axetil [1]	0.5 x 2	5.2 ± 1.0	30-50	1.1-1.4	8	0

Common dosages (dose x number of doses per day),

MICs are based on the epidemiological cut-off values (ECOFFs) from the EUCAST MIC distributions

References

- [1] Nix et al., Pharmacotherapy 1997; 17:121-125
- [2] La Rosa et al., Antimicrob Agents Chemother 1982; 320-322
- [3] Ding et al., Br J Clin Pharmacol 2011;422-427.
- [4] Faulkner et al., J Antimicrob Chemother 1988; 21:787-794
- [5] Borin et al., J Clin Pharmacol 1994; 34:774-781
- [6] Lin et al., Antimicrob Agents Chemother 1995; 39:356-358